



Tutorial 2C: Vectors III – Equations of Planes

Section A (Basic Questions)

- 1 (a)** The Cartesian equation of the plane π is $3x - y + 4z = -2$. State the equation of π in scalar product form.
- (b)** The equation of the plane π is $\mathbf{r} \cdot (-\mathbf{i} + \mathbf{j} + 2\mathbf{k}) = 4$. Write down the Cartesian equation of π , and find a vector equation of π , involving two parameters.
- (c)** The vector equation of the plane π is $\mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} + s \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$, $s, t \in \mathbb{R}$. Find the equation of π in both the scalar product form and the Cartesian form.
- [**(a)** $\mathbf{r} \cdot (3\mathbf{i} - \mathbf{j} + 4\mathbf{k}) = -2$, **(b)** $-x + y + 2z = 4$, $\mathbf{r} = (-4\mathbf{i}) + \lambda(\mathbf{i} + \mathbf{j}) + \mu(2\mathbf{i} + \mathbf{k})$, $\lambda, \mu \in \mathbb{R}$
(c) $\mathbf{r} \cdot (\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) = -4$, $x + 2y + 2z = -4$]

Solution

(a)

$$\pi : \quad 3x - y + 4z = -2 \quad \text{- Cartesian equation}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = -2$$

- scalar product form

$$\mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} = -2$$

(b) $\pi : \quad \mathbf{r} \cdot \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = 4 \quad \text{- scalar product form}$

$$-x + y + 2z = 4 \quad \text{- Cartesian equation}$$

So $x = y + 2z - 4$.

$$\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -4 + y + 2z \\ y \\ z \end{pmatrix}$$

Letting $\lambda = y$ and $\mu = z$, a vector equation of the plane π is

$$\mathbf{r} = \begin{pmatrix} -4 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \quad \lambda, \mu \in \mathbb{R} \quad \text{- vector equation}$$

(c) $\pi: \mathbf{r} = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} + s \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, s, t \in \mathbb{R}$ - vector eqn

$\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$ are both parallel to π .

A vector perpendicular to π is $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -2 \end{pmatrix}$.

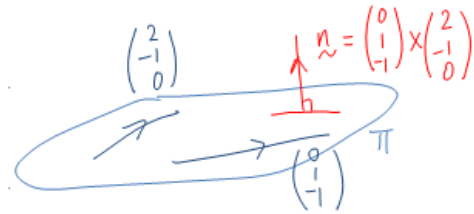
Hence, the required equation of the plane is

$$\mathbf{r} \cdot \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = -4,$$

i.e. $\mathbf{r} \cdot \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = -4$ - scalar product form

Cartesian equation :

$$x + 2y + 2z = -4$$



2 For each of the following cases, find the equation of the plane π_1 in scalar product form.

(a) π_1 contains the point $A(2, -4, 7)$ which is the foot of perpendicular from the origin to π_1 .

(b) π_1 contains the point $B(1, -2, 0)$ and is parallel to the plane $\pi_2: 2x - y + 3z = -1$.

(c) π_1 passes through the points $C(2, 3, 3)$, $D(0, 1, 0)$, and $E(-1, 5, 1)$.

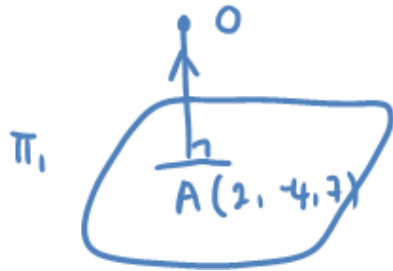
(d) π_1 contains the lines $l_1: x+3=y-1, z=-1$ and $l_2: \mathbf{r} = (-3-t)\mathbf{i} + \mathbf{j} - (1+t)\mathbf{k}, t \in \mathbb{R}$.

(e) π_1 contains the line $l: \mathbf{r} = (-2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}) + \lambda(2\mathbf{i} + \mathbf{j} + 2\mathbf{k}), \lambda \in \mathbb{R}$ and is perpendicular to the plane $\pi_2: \mathbf{r} \cdot (7\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}) = 2$.

(f) π_1 passes through the points $F(1, 1, 1)$ and $G(3, 2, 2)$ and is perpendicular to the plane π_2 that is parallel to the vectors $-\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + \mathbf{k}$.

[(a) $\mathbf{r} \cdot (2\mathbf{i} - 4\mathbf{j} + 7\mathbf{k}) = 69$ (b) $\mathbf{r} \cdot (2\mathbf{i} - \mathbf{j} + 3\mathbf{k}) = 4$ (c) $\mathbf{r} \cdot (2\mathbf{i} + \mathbf{j} - 2\mathbf{k}) = 1$

(d) $\mathbf{r} \cdot (-\mathbf{i} + \mathbf{j} + \mathbf{k}) = 3$ (e) $\mathbf{r} \cdot (-3\mathbf{i} + 4\mathbf{j} + \mathbf{k}) = 23$ (f) $\mathbf{r} \cdot (-\mathbf{i} + \mathbf{j} + \mathbf{k}) = 1$]

Solution**(a)** $\overline{OA}$ is $\perp$ to π_1

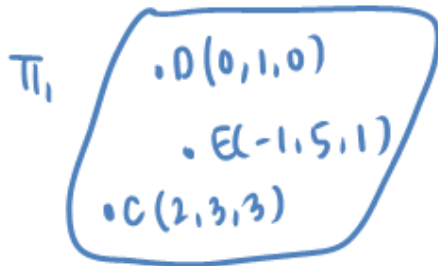
$$\mathbf{r} \cdot \begin{pmatrix} 2 \\ -4 \\ 7 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ 7 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -4 \\ 7 \end{pmatrix} = 69,$$

$$\text{i.e. } \mathbf{r} \cdot \begin{pmatrix} 2 \\ -4 \\ 7 \end{pmatrix} = 69$$

(b)

$$\mathbf{r} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = 4,$$

$$\text{i.e. } \mathbf{r} \cdot \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = 4$$

(c)

$$\mathbf{r} = \overline{OC} + \lambda \overline{CD} + \mu \overline{CE}$$

$$= \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -2 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} -3 \\ 2 \\ -2 \end{pmatrix}, \lambda, \mu \in \mathbb{R}$$

A vector perpendicular to π is $\begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \\ -10 \end{pmatrix}.$

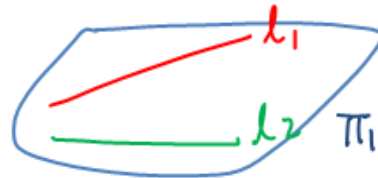
$$\mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = 1, \text{ i.e. } \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = 1$$

(d) $l_1: x+3=y-1, z=-1$

$$l_1: \mathbf{r} = \begin{pmatrix} -3 \\ 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \lambda \in \mathbb{R} \quad \text{and} \quad l_2: \mathbf{r} = \begin{pmatrix} -3 \\ 1 \\ -1 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}, t \in \mathbb{R}$$

Since π_1 contains the lines l_1 and l_2 , a vector equation of π_1 is

A vector perpendicular to π_1 is $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$



Hence

$$\mathbf{r} \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = 3, \quad \text{i.e. } \mathbf{r} \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = 3$$

(e)

Since π_1 is $\perp$ to π_2 , π_1 is $//$ to $\mathbf{n}_2 = \begin{pmatrix} 7 \\ 4 \\ 5 \end{pmatrix}$

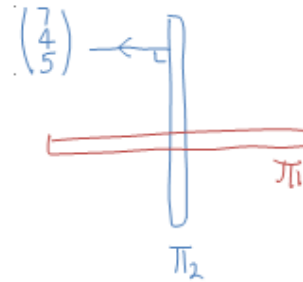
Hence, a vector equation of π_1 is $\mathbf{r} = \begin{pmatrix} -2 \\ 5 \\ -3 \end{pmatrix} + \alpha \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} + \beta \begin{pmatrix} 7 \\ 4 \\ 5 \end{pmatrix}, \alpha, \beta \in \mathbb{R}$

A vector perpendicular to π_1 is $\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 7 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix}$.

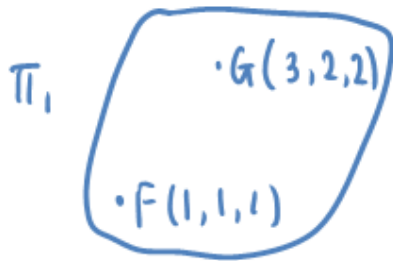
Hence

$$\mathbf{r} \cdot \begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \\ -3 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix} = 23,$$

$$\text{i.e. } \mathbf{r} \cdot \begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix} = 23$$



(f)



Since $\pi_1 \perp \pi_2$, π_1 is parallel to $\mathbf{n}_2$.

$$\mathbf{n}_2 \text{ is parallel to } \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 9 \\ -3 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

$$\text{Let } \mathbf{n}_2 = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

A vector perpendicular to π_1 is

$$\overrightarrow{FG} \times \mathbf{n}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} -4 \\ 4 \\ 4 \end{pmatrix}$$

$$\text{Hence, } \mathbf{r} \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = 1,$$

$$\text{i.e. } \mathbf{r} \cdot \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} = 1$$

3 Show that the line $l: x = \frac{y-7}{7} = \frac{z-5}{2}$ is contained in the plane $\pi: 3x - y + 2z = 3$.

$$l: \quad \mathbf{r} = \begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 7 \\ 2 \end{pmatrix}, \lambda \in \mathbb{R} \quad \text{and} \quad \pi: \quad \mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 3$$

Any point on l has position vector $\begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 7 \\ 2 \end{pmatrix}$ for some $\lambda \in \mathbb{R}$.

$$\left[\begin{pmatrix} 0 \\ 7 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 7 \\ 2 \end{pmatrix} \right] \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = (0 - 7 + 10) + \lambda(3 - 7 + 4) = 3$$

So any point on l is also on π

Thus l is contained in π .